

COMPX361-22B Logic and Computation

Assignment 2

Due 4PM, Friday 19 August (end of Week 5)

1. [6 marks.] Consider the DFSA $M = (Q, \delta, q_0, F)$, where the input alphabet is the binary alphabet $X = \{0, 1\}$, $Q = \{q_0, q_1, q_2, q_3\}$, with q_0 the start state, $F = \{q_3\}$ the set of accepting states, and δ is defined by the following table:

δ	0	1
q_0	q_1	q_2
q_1	q_3	q_2
q_2	q_1	q_3
q_3	q_0	q_0

After first drawing the state diagram for M , determine a regular expression for $L(M)$. Use the transition graphs method given in lectures, showing all your steps carefully. (You will need some unused name for your new start state, such as Q_0 . Also, you may like to re-name some regular expressions that arise in your method with names such as $R_1, R_2, \dots$, to make it easier to proceed. If so, make it clear how you are doing this, so that your final answer is meaningful!)

2. [1 mark.] Let $R = (abc^* + a + ab)c^* + (bc)^*$. Write down a regular expression for $R^\leftarrow$.
3. [7 marks.] The DFSA $M = (Q, \delta, q_0, F)$ has input alphabet $X = \{0, 1\}$, state set $Q = \{q_0, q_1, q_2, q_3\}$, with q_0 the start state, $F = \{q_0, q_1, q_2\}$ the set of accepting states, and δ defined by the following table:

δ	0	1
q_0	q_1	q_3
q_1	q_2	q_3
q_2	q_2	q_3
q_3	q_3	q_3

Draw M , then find an NFSA and then a DFSA that accepts $L(M)^\leftarrow$ (using the dual machine construction and then the subset method), and repeat to find a minimal DFSA equivalent to M . Just give its state diagram.

4. [5 marks.] Prove that the binary language $\{0^n 10^n \mid n \geq 0\}$ is not regular, using the form of the Pumping Lemma given in class.

(turn over)

5. [3+3+1 marks.] A Turing machine $M = (Q, \delta, q_0, q_F)$ has input alphabet $X = \{x, y, z\}$, tape alphabet $Y = \{x, y, z, A, b\}$ (where “ b ” is “blank”), $Q = \{q_0, q_1, q_2, q_3, q_4, q_F\}$, and the transition (partial) function δ given by the following table:

δ	x	y	z	A	b
q_0	(q_1, A)	(q_1, A)	(q_1, A)	-	(q_F, b)
q_1	-	-	-	(q_2, R)	-
q_2	(q_3, b)	(q_3, b)	(q_3, b)	-	(q_4, L)
q_3	-	-	-	-	(q_2, R)
q_4	-	-	-	(q_F, b)	(q_4, L)

- Draw the state diagram for M .
 - Give the configuration sequence for the input string xyz .
 - What do you think this Turing machine is doing in general? For a given input, what is the output? Give a clear simple description in one sentence.
6. [4 marks.] Design a Turing machine that takes as input any binary string and decides if the input string is empty: make it give output “1” if it is, and “0” if not. Remember that *only* these (short) strings should remain on the tape. Just give the state diagram. *No flag symbols allowed*, so the tape alphabet must equal the input alphabet $\{0, 1\}$ plus “ b ” only.